Vaibhav Maheshwari

Full Stack Software Engineer

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Professional Summary

Full Stack Software Engineer with experience building scalable goal-tracking platforms at D.E. Shaw using Node.js, React, MongoDB, and Redis. Shipped production AI features including LLM-powered check-in suggestions. Strong in performance optimization (SSR, Brotli, caching), event-driven architecture (Apache Kafka), RESTful API design, and React UI engineering—with measurable impact on latency, adoption, and user experience.

Technical Skills

Languages: JavaScript / TypeScript (Primary), Python, SQL, Java
Frameworks: Node.js, Express.js, React, RESTful APIs, Jest, Microservices
Databases: MongoDB (aggregation, schema design), Redis (caching), SQLite, PostgreSQL
AI & LLM Tools: Gemini AI, OpenAI API, Google ADK, prompt engineering, LLM tool calling
Distributed Systems: Apache Kafka, Redis pub/sub, event-driven architecture, async job processing
Performance: Server-side rendering (SSR), Brotli compression, query optimization, memoization
Tools & Cloud: Git, Docker, CI/CD, Agile, Linux

Work Experience

The D.E. Shaw Group   **Hyderabad, India**
Software Engineer (Full Stack) — DESGoals   Aug 2024 – Present

Node.js, React, TypeScript, MongoDB, Redis, Kafka

Internal OKR and goal-tracking platform enabling teams to set objectives, track progress, and run structured check-ins across the organization.

- Built a Redis caching layer for read-heavy goal dashboards, reducing p95 read latency from 340ms to under 95ms (72% improvement) and serving 2,000+ active users with sub-100ms reads on cached goal summaries.
- Implemented server-side rendering with Brotli compression for goal summary pages, cutting Time-to-Interactive from 2.8s to 1.1s (61% improvement) and reducing payload size by 48% on initial page load.
- Leading migration of the goal check-in notification pipeline to Apache Kafka, decoupling write events from API handlers—improving write-to-cache propagation to under 1 second for 95% of check-in updates and enabling parallel consumer scaling.
- Engineered React dashboards with virtualized lists, memoized selectors, and lazy-loaded modules—reducing unnecessary re-renders by 55% on large OKR tree views and improving scroll performance on dashboards with 500+ goals.
- Designed RESTful Node.js APIs backed by MongoDB aggregation pipelines for goal CRUD, check-ins, and progress rollups—maintaining 99.9% uptime during quarterly OKR cycles.
- Optimized MongoDB query patterns and added compound indexes on frequently filtered fields (owner, quarter, status)—cutting slow-query volume by 40% and improving write-path throughput during peak check-in windows.
- Built reusable React component library for forms, filters, and data tables with consistent loading/error states—reducing new dashboard delivery time by 30% and improving Lighthouse accessibility scores from 78 to 94.

AI Products Shipped (Production)

AI-Powered Check-in Suggestions   **D.E. Shaw (DESGoals)**
Node.js + Gemini AI + Prompt Engineering   *2024 – Present*

- Built and shipped a production LLM suggestion engine for goal check-ins: ingests historical OKR progress, team context, and prior check-in notes to generate contextual prompts via Gemini—helping users complete check-ins faster with higher quality inputs.
- Reduced average check-in completion time by 35% (from 4.2 min to 2.7 min) and increased weekly check-in adoption by 22% across pilot teams within one OKR cycle.
- Processing 500+ AI suggestion requests/day in production with 99.5% service uptime; achieved 88% user acceptance rate on suggested prompts without manual edits.
- Designed prompt templates with guardrails for tone, brevity, and data privacy—ensuring no sensitive goal data leaked into model logs while maintaining suggestion relevance.

Projects

SmartCart AI — Multi-Platform Grocery Price Comparator

Python (FastAPI), React, LLM APIs, Web Scraping *Personal Project*

- Built an AI-powered grocery cart assistant that accepts natural-language input (e.g., "2L Amul milk, brown bread, 1kg rice") and parses items using LLM-based extraction with 92% field accuracy across brand, quantity, and unit.
- Aggregates real-time prices from multiple grocery platforms (Blinkit, Zepto, BigBasket, Instamart), compares total cart cost side-by-side, and highlights the cheapest option with item-level price breakdowns.
- Flags items unavailable on specific platforms ("not listed" detection), suggests closest substitutes, and alerts users when a platform lacks 20%+ of cart items—reducing failed checkout attempts in testing.
- Delivered a React dashboard with cart builder, platform comparison table, and savings summary—helping a 50-user test cohort save an average of 18% on weekly grocery spend.

Education

Indian Institute of Information Technology Allahabad (IIIT-A) **Prayagraj, India**

Bachelor of Technology in Information Technology (CGPA: 8.36/10.0)

Relevant Coursework: Operating Systems, Database Management Systems, Data Structures & Algorithms, Computer Networks

Achievements

- Codeforces — **kAB-d** (Specialist)
- LeetCode — **Ironman15**
- CodeChef — **vabsrts15** (4-Star)